

ChE 433 – Process Simulation with ASPEN Plus –Fall 2019

Student Learning Outcome 2

Rubric for assessment problem on Homework Problems 4 and 5

- 4A .Summary of results for both problems (about one page)
- 4B. Flowsheets for both processes
- 4C. Heat and Weight Balance
- 4D. List of Equipment with the capital costs (see example below)
- 4E. List of Utilities/Raw Material/Product with operating costs

- 4A. Outline your steps you followed to reach your final column design
- 4B. Final process flowsheet printout
- 4C. ASPEN report for the distillation column
- 4D. Heat and Weight Balance for the entire unit
- 4E. List of all equipment for with the capital costs (see example below)
- 4F. List of utilities, raw materials and products with operating costs

SLO 2 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Developing a design strategy Problem 4A,4B,5B,5D 100 pts	No design strategy; haphazard approach (<70 pts)	Uses a design strategy with guidance (70-90 pts)	Develops a design strategy, decomposition of work into subtasks, development of a timetable (>90 pts)
2. Applying engineering and/or scientific principles correctly to design practical processes Problem 4C,4D,4E,5D 100 pts	No application of engineering and/or scientific principles (<70 pts)	Includes only minor or cursory consideration of economic, safety, and environmental constraints (70-90 pts)	Applies engineering and/or scientific principles correctly to design practical processes (>90 pts)
5. Supporting design procedure with documentation and references Problem 5A,5C 100 pts	Design is done incompletely without the proper equations and without references (<70 pts)	Minimal or incorrect use of computer tools and engineering resources (70-90 pts)	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem (>90 pts)

Rubric					
PI	Problem	Total pts	Unsatisfactory	Developing	Satisfactory
1	4A,4B,5B,5D	100	<70	70-90	>90
2	4C,4D,4E,5D	100	<70	70-90	>90
5	5A,5C	100	<70	70-90	>90

RESULTS BY PERFORMANCE INDICATOR			
Rubric #	1	2	5
Mapped SO	2	2	2
Average	2.26	2.87	2.64
Std Dev.	0.64	0.34	0.49
S	14	34	25
D	21	5	14
U	4	0	0
Checksum	39	39	39

RESULTS BY STUDENT OUTCOME			
SO	2		
Average Score	2.59		
Standard dev.	0.25		

Average scores were fairly uniform although there was some variation among students. More students were strong in applying engineering and/or scientific principles correctly to design practical processes. Overall performance was satisfactory for all performance indicators despite the very stringent criteria used for assessment.

Problem 4: Cyclohexane production

For the cyclohexane production process of **Assignment #3**, reconfigure the flowsheet using unit operations that correspond to actual equipment in the field. For heat exchangers only, if you are going to use any, use two heaters as demonstrated in classroom lectures. Prepare a list of all equipment (except the reactor) and calculate the total capital cost. Prepare a list with all utilities and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. Starting with your own simulation from **Assignment #3**. Plant capacity is 300 KMTA of cyclohexane with plant operating data defined in **Assignment #3**. Cooling water is available by a nearby cooling tower that has been designed for dry bulb temperature of 85 deg F, relative humidity of 70% and an approach temperature of 10 deg C. The maximum hot water temperature was taken as 45 deg C for the design of the cooling tower. Multiple pressure levels of steam are available by a steam plant nearby.

REQUIRED REPORT SECTIONS

- A. Summary of results for both problems (about one page)
- B. Flowsheets for both processes
- C. Heat and Weight Balance
- D. List of Equipment with the capital costs (see example below)
- E. List of Utilities/Raw Material/Product with operating costs

1	2	A	B	C	D	E	F	G
	1	Cyclohexane Production Cost Estimate						
	2	EQUIPMENT LIST: CAPITAL COSTS						
	3		<i>Project:</i>	Assignment #2	<i>Chemical Engineers:</i>			
	4		<i>Location:</i>	Chemical Engineering Department	<i>Date:</i>			
	5	Equipment Class	ASPEN BlockID	Service Description	Tag	Material Cost	Installed Cost	Annual Cost, \$/yr
	6	Distillation						
	7	Towers	xxxx	Finishing Column	T-1	\$1,950,958	\$4,095,200	\$819,040
	8	Towers	xxxx	Packing, Trays, Internals	T-1			
	9	Drums						
	10	Vertical Tank	xxxx	Separation Drum	V-1			
	11	Vertical Tank	xxxx	Reflux Drum	V-2			
	12	Heat Exchangers						
	13	S&T Exchanger		Feed-Product Exchanger	E-1			
	14	S&T Exchanger		Product Cooler	E-2			
	15	Air Coolers						
	16	Air Cooler		Reflux Condenser	EA-1			
	17	Air Cooler			EA-2			
	18	Pumps						
	19	Pump		Reflux Pump	P-1			
	20	Pump			P-2			
	21	Turbines						
	24	Compressors						
	27	Reactors						
	30							

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I - EXECUTIVE SUMMARY:

A. Cyclohexane Production

The cyclohexane production schematic from previous homework was reconfigured using the real-life equipment. The objective of this homework was to produce 300 KMTA cyclohexane via hydrogenation of benzene in bubble column reactor in the presence of nickel catalyst. The reaction mixture is fed to the reactor to convert the raw materials into the cyclohexane where the conversion rate of the reactor is given to be 99.5 %. The hydrogen stream is introduced at 335 psia. Benzene feed is initially at 15 psia therefore to equate the pressures of the stream entering the mixer, a pump is used to pressurize the benzene feed up to 335 psia. The reactor inlet is preheated using the heater duty at the outlet of the reactor and acts as shell and tube heat exchanger. To cool down the reactor effluent, cooling water is used from the cooling tower designed for dry bulb temperature of 85 F. Using psychrometric chart, wet bulb temperature is identified, and temperature of the cooling water was calculated which was found to be 95 F. The cooled stream is then introduced in a high-pressure separator where gas phase obtained from the top is recycled to feed of the reactor. This recycled stream is compressed to the pressure to 335 psia in order to be introduced into the reactor along with hydrogen and benzene feed. Also, to avoid accumulation of unreacted components at the top of the high-pressure vessel, the purge is carried out of the system. The liquid phase obtained from the bottom of the separation vessel is split into two streams where one is recycled back to the reactor and the other is introduced into the low-pressure separating vessel. The recycled liquid stream is pressurized to 335 psia to be able to merge into the reaction mixture. Hence, to process cyclohexane in an actual chemical plant we will need to install a shell and tube heat exchanger system, a cooler, two pumps and a compressor. After running the simulation and doing cost analysis, the total capital cost was found to be \$7,723,793. Total operating cost and the cost of raw materials for the utilities was found to be \$7,577,079 and \$3,186,005,18 respectively.

II – SYSTEMS AND PROCEDURES:

PROBLEM A: A schematic of refrigeration with flash tank economizer and two stage compressors was provided in the handout was utilized to solve this problem. The purpose of this problem was to simulate a three-stage compressor refrigeration system which was done by adding another set of a compressor, separator and a valve. The given physical properties were inserted in ASPEN simulation.

Using **Transfer tool** in ASPEN, the pressures from compressor 1, compressor 2 and compressor 3 were transferred to outlet pressures of Valve 2, Valve 1 and S3 respectively. **Two Design Specs** were also used. Chiller's power was defined to not be more than 3 MW by varying the molar flow rate of the stream S10 which was taken to be the feed basis of 100 kmol/hour. Second, design spec was made by defining outlet pressure variables, VALVOUT and CHILLERO of Valve 3 and Chiller, and the difference between them was targeted to be equal to zero by varying outlet pressure of Valve 3. This assure the pressure across the chiller to be constant.

Optimization, in Model Analysis Tools, was performed to obtain the required total power as a function of interstage pressure. Variables, W1, W2, and W3 which are power of each compressor, were defined in Optimization tab and the sum of all three powers were minimized in Objective & Constraint tabs. The outlet pressures of the Valve 1 and Valve 2 were varied from 1 bar to 7 bar and 7 bar to 12 bar respectively, to meet the specified objective. The range of the pressures was guessed based on the known pressure range around the Valve 1 of the system from Assignment #3, i.e. 5.07 bar to 12 bars The pressure guessed for the second stage was calculated by,

$$P_2 = \sqrt{P_1 * P_3}$$

$$P_2 = \sqrt{5.07 * 12.05} = 7.75 \text{ bar}$$

A Calculator was created to obtain the Coefficient of Performance (COP). The Powers of all the compressors, the duty of the Chiller and COP were defined as export variables. The calculation formulas were specified in Calculate/FORTRAN.

PROBLEM B: To run the simulation, two **Design Specifications** were defined. First, for the tear stream, for the present composition of hydrogen and benzene in the tear namely TEARH and TEARB. In the **Spec** tab the ratio of TEARH and TEARB was targeted to be equal to 3.3 by **Varying** the flow fraction of the splitter B4. **Second design specification** was defined for the product stream (PRODUCT) where the mass flow rate of the product was targeted to be equal to **300** by **Varying** the benzene feed (BZFEED) molar flow rate. The molar flow of hydrogen was reevaluated after the design spec was set up using the **Calculator** in ASPEN. Since the recycle liquid should not be more than 45 % of the benzene feed, the ratio of liquid recycles (RECLIQ) and the benzene feed (FEEDB) was made equal to 0.45 again using **Calculator**. New variables **FEEDB, HNEW, and RECLIQ** were defined in order to make the Fortran calculation.

III – RESULTS:

PROBLEM A: The results obtained of ASPEN optimization gave a COP of 2.16 for Total work of 1.39 MW.

Table 1: Compressors' Work

Sampled variable	Initial value	Final value	Units
W1	0.069555263	0.725891269	MW
W2	0.0355739719	0.3022339	MW
W3	0.34703546	0.359681847	MW
TOTAL Work		1.38780702	MW

Table 2: Interstage Pressures and Coefficient of Performance

Name	Value	Units
INTERSTAGE PRESSURE 1	5.07393789	Bar
INTERSTAGE PRESSURE 2	7.7608866	Bar
COP	2.16168354	

Table 3: Equipment used for cyclohexane production with the capital costs

Cyclohexane Production Cost Estimate				
EQUIPMENT LIST: CAPITAL COSTS				
	Assignment #2			
	Chemical Engineering Department			
Equipment Class	Service Description	Material (equipment) Cost	Installed Cost	Annual Cost, \$/yr
Drums				
Vertical Tank	High Pressure Drum	\$81,426	\$325,703	\$434,271
Vertical Tank	Low Pressure Drum	\$33,543	\$134,171	\$178,894
Heat Exchangers				
S&T Exchanger	Feed-Product Exchanger	\$314,129	\$1,099,450	\$1,465,933
Water Coolers				
Water Cooler		\$53,638	\$187,732	\$250,310
Pumps				
Pump1	Pump used to pressurize benzene feed	\$13,869	\$55,476	\$73,968
Pump2	Pump used fo recycle liquid	\$12,885	\$51,541	\$68,722
Compressors				
Compressor	Recycle Gas Compressor	\$1,531,495	\$3,828,736	\$5,104,982

Table 4: Utilities, Raw Material, Product with operating costs

Utility Class					
ELECTRICITY	power (kW)	COST(\$/kW)	HOURS	YEARLY COST (\$)	
PUMP1	30.841	0.0657	8640	17506.83197	
PUMP2	1.295	0.0657	8640	735.10416	
COMPRESSOR	22.376	0.0657	8640	12701.69165	
TOTAL				30943.62778	
COOLING WATER	Power (kW)	\$/m3	LMTD (K)	Specific Heat Capacity (kJ/kg.s)	Yearly Cost
	669.805	0.065	76.654	4.187	4277.899381
RAW MATERIALS	\$/GAL	GAL/HR	HR/YR	\$/YR	
BENZENE	2.75	9970	8640	236887200	
	\$/KG	KG/HR	HR/YR	\$/YR	
HYDROGEN	3	3152.52	8640	81713318.4	
PRODUCTION	\$/TON	TON/YR		\$/YR	
CYCLOHEXANE	1900	300000		570000000	

Table 5: Raw Materials and Product

Parameters	Value	Units
Product	300	ktonne/year
Benzene Feed	915.23	lbmol/hr
Hydrogen Feed	2864.68	lbmol/hr

Table 6: Total Costs

Total Capital Cost	Total Operating Cost	Total Raw Material Cost	Product Revenue	Fuel Credits
\$7,723,793	\$7,577,079	\$3,186,005,18.4	\$570,000,000	\$2393511.12

IV – DISCUSSION:

V – CONCLUSIONS:

PROBLEM B: The aim of this project was to simulate the actual production plant to produce cyclohexane using the bubble column reactor. The reactor used had the 99.5 % conversion. By focusing on the accurate utilities, a shell and tube heat exchanger system, a cooler, two pumps and a compressor were installed to process cyclohexane in a bubble column reactor. The total capital cost of equipment was calculated by utilizing the equations that calculate the cost of the equipment by taking in account of the indexes of the year 2010 and 2020 for a good estimation. The value of the total capital cost was found to be \$7,723,793. The total operating cost was also evaluated utilizing the annual cost required for the maintenance of the equipment which was found to be \$7,577,079. The total cost of the raw materials was evaluated to find out how much money should be spent on buying raw materials to produce 300 KMTA cyclohexane. The raw material cost was found to be \$318,600,518.

VI – APPENDICES:

APPENDIX I:

PROBLEM B:

For Design Specifications in ASPEN Plus we used,

$$\frac{TEARH}{TEARB} = 3.3$$

$$HNEW = 3.13 * FEEDB \text{ (SOLVER ASPEN)}$$

Calculating equipment, installation, and annualized cost

For Shell and Tube Heat Exchanger and a Cooler:

$$\$ (2010) = 28000 + 54 (A[m^2])^{1.2}$$

$$\$ (2020) = \$ (2010) * \frac{\text{index } (\$ 2020)}{\text{index } (\$ 2010)}$$

Where A can be found using,

$$Q = F_T A U \Delta T_{LN}$$

Where, ΔT_{LN} is given as,

$$\Delta T_{LN} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

The installation factor for the shell and tube heat exchanger is given to be 3.5 therefore,

$$\text{Installation cost} = 3.5 * \text{purchasing cost}$$

The annualized cost can be found using,

$$\text{Annualized cost} = \frac{4}{3} * \text{installation cost}$$

For Pumps

$$\$ (2010) = 8000 + 240 \left(Q \left[\frac{L}{s} \right] \right)^{0.9}$$

Installation factor is given to be 4.0. Also, the installation cost and annualized cost can be found from the same formulae.

For Compressor

$$\$ (2010) = 58000 + 20000 (W[kW])^{0.6}$$

Installation factor is given to be 2.5. Also, the installation cost and annualized cost can be found from the same formulae.

For Drums (Vertical Vessel)

$$\$ (2010) = 11600 + 34 (M[kg])^{0.85}$$

Where M is given as,

$$M[kg] = \pi D_i L t \rho_{STEEL}$$

Where t is the thickness of the walls of the drum and is given as,

$$t = \frac{P_i D_i}{2 S E - 1.2 P_i}$$

Where,

D_i is the internal diameter

P_i is the internal pressure of the drum + 10 % safety factor

E = welded efficiency

S = Stress material can take [=] ksi

Where,

$$I \text{ ksi} = 6.895 \frac{N}{m^2}$$

Internal Diameter can be calculated using the residence time,

$$\tau = \frac{V}{F}$$

Where volume of the vessel is given as,

$$V_{VESSEL} = 2 V$$

And the length is given as,

$$L = 2 D_i$$

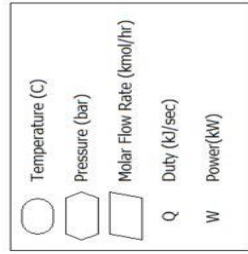
Where,

$$D_i = \left(\frac{2 V_{VESSEL}}{\pi} \right)^{\frac{1}{3}}$$

Installation factor is given to be 4.0. Also, the installation cost and annualized cost can be found from the same formulae.

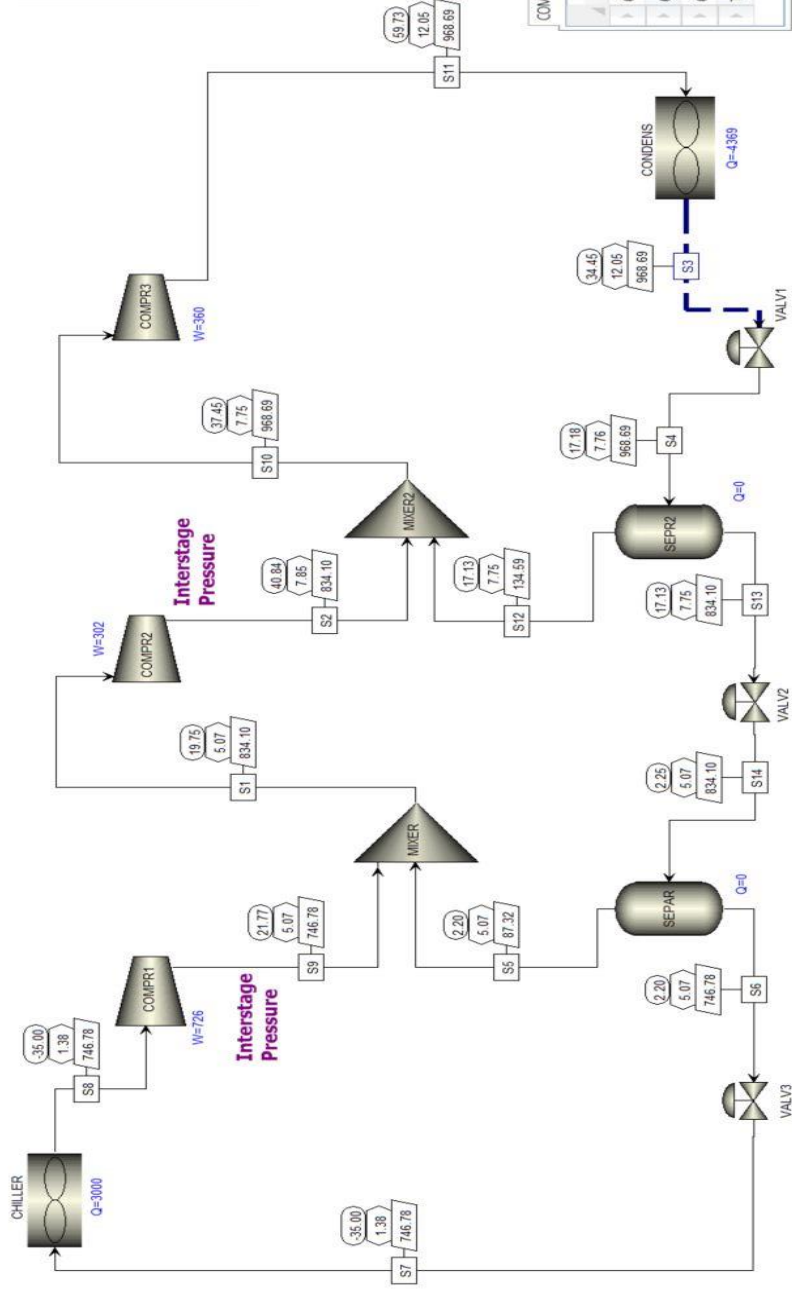
Figure 1: Optimized Refrigeration Process Flow Diagram

APPENDIX II:



INTERSTAGE PRESSURES	
Name	Value
INTERSTAGE PRESSURE 1 (BAR)	5.07394
INTERSTAGE PRESSURE 2 (BAR)	7.76089
COP	2.16168

COMPRESSORS	
Name	Value
COMPR 1 WORK (MM)	0.725891
COMPR 2 WORK (MM)	0.302234
COMPR 3 WORK (MM)	0.359682
TOTAL WORK (MM)	1.38781



APPENDIX III:



Problem A: Stream Summary - I

Material									
Stream Name	Units	S1	S2	S3	S4	S5	S6	S7	
Description									
From		MIXER	COMPR2	CONDENS	VALV1	SEPAR	SEPAR	VALV3	
To		COMPR2	MIXER2	VALV1	SEPR2	MIXER	VALV3	CHILLER	
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	
Maximum Relative Error						4.05E-11			
MIXED Substream									
Temperature	C	19.746	40.8401	34.4519	17.1821	2.19602	2.19602	-34.9993	
Pressure	bar	5.06625	7.85269	12.0466	7.76089	5.06625	5.06625	1.3766	
Mass Liquid Fraction		0	0	1	0.861422	0	1	0.786436	
Mass Solid Fraction		0	0	0	0	0	0	0	
Mass Enthalpy	kJ/kg	-2396.68	-2367.1	-2712.54	-2712.54	-2426.63	-2800.5	-2800.5	
Mass Entropy	kJ/kg-K	-6.47098	-6.44728	-7.63711	-7.62944	-6.57642	-7.93424	-7.90483	
Mass Density	gm/cc	0.0101161	0.0150949	0.476884	0.100181	0.01099	0.525927	0.014686	
Average MW		44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	
Mole Fractions									
PROPANE		1	1	1	1	1	1	1	
Mass Fractions									
PROPANE		1	1	1	1	1	1	1	
Mass Flows	kg/hr	36780.8	36780.8	42715.9	42715.9	3850.6	32930.2	32930.2	
PROPANE	kg/hr	36780.8	36780.8	42715.9	42715.9	3850.6	32930.2	32930.2	
Mass Vapor Fraction		1	1	0	0.138578	1	0	0.213564	

Problem A: Stream Summary - II

Material													
Stream Name	Units	S8	S9	S10	S11	S12	S13	S14					
Description													
From		CHILLER	COMPR1	MIXER2	COMPR3	SEPR2	SEPR2	VALV2					
To		COMPR1	MIXER	COMPR3	CONDENS	MIXER2	VALV2	SEPAR					
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN					
Maximum Relative Error		3.21E-11				-2.05E-10							
MIXED Substream													
Temperature	C	-35	21.765	37.4546	59.7261	17.1301	17.1301	2.24649					
Pressure	bar	1.37656	5.06625	7.75	12.0466	7.75	7.75	5.07394					
Mass Liquid Fraction		0	0	0	0	0	1	0.895633					
Mass Solid Fraction		0	0	0	0	0	0	0					
Mass Enthalpy	kJ/kg	-2472.53	-2393.18	-2373.06	-2342.74	-2409.98	-2761.36	-2761.36					
Mass Entropy	kJ/kg-K	-6.52768	-6.45907	-6.46418	-6.44128	-6.58711	-7.79759	-7.79212					
Mass Density	gm/cc	0.00320078	0.010026	0.0151012	0.022894	0.016725	0.504424	0.0893998					
Average MW		44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0965					
Mole Fractions													
PROPANE		1	1	1	1	1	1	1					
Mass Fractions													
PROPANE		1	1	1	1	1	1	1					
Mass Flows	kg/hr	32930.2	32930.2	42715.9	42715.9	5935.16	36780.8	36780.8					
PROPANE	kg/hr	32930.2	32930.2	42715.9	42715.9	5935.16	36780.8	36780.8					
Mass Vapor Fraction		1	1	1	1	1	0	0.104367					

Problem B: Stream Summary - I

	BZFEED	EFFLUENT FUEL	H2FEED	HPBOT	HPOVHD	PRODUCT PURGE	RECGAS
Substream: MIXED							
Mole Flow lbmol/hr							
H2	0	263.7353	2.920142	4.202969	259.5299	0.021936	31.74449
N2	0	97.22058	2.882907	4.209041	93.01113	0.063422	11.37669
C1	0	400.8424	28.27491	42.8348	358.0032	1.709456	43.78929
BZ	915.2347	1.831593	0.021543	0	1.810608	1.245882	0.002567
CH	0	1313.536	14.49015	0	1299.241	894.9785	1.748455
Mole Frac							
H2	0	0.126969	0.060098	0.975	0.003108	0.358042	0.358042
N2	0	0.046804	0.059332	0.005	0.003113	0.128316	0.128316
C1	0	0.192976	0.581912	0.02	0.031676	0.493893	0.001904
BZ	1	0.000882	0.000443	0	0.001339	2.89E-05	0.001387
CH	0	0.632369	0.298215	0	0.960765	0.019721	0.996614
Total Flow lbmol/hr	915.2347	2077.166	48.58965	2864.684	1352.298	724.8599	88.6615
Total Flow ktonne/year	284.2671	478.6446	7.00385	27.63805	438.577	40.06728	300
Total Flow l/sec	10.48398	410.3185	156.0469	423.9103	18.78906	114.211	12.64936
Temperature F	100	400	115.1254	120	120	120	115.1254
Pressure psia	15	320	15	335	310	310	15
Vapor Frac	0	1	1	1	0	1	0
Liquid Frac	1	0	0	0	1	0	1
Solid Frac	0	0	0	0	0	0	0
Enthalpy J/kmol	50944100	-7.6E+07	-7.9E+07	-777550	-1.5E+08	-3.9E+07	-1.5E+08
Enthalpy J/kg	652179	-1303500	-2183000	-320450	-1823600	-2785400	-1815200
Enthalpy Watt	5874750	-2E+07	-484490	-280650	-2.5E+07	-3536600	-1.7E+07
Entropy J/kmol-K	-246580	-316800	-190150	-24322.7	-581740	-64436	-601430
Entropy J/kg-K	-3156.67	-5466.58	-5245.33	-10024.2	-7132.23	-4635.11	-7158.43
Density kmol/cum	10.99942	0.637842	0.039233	0.851464	9.068403	0.799667	8.945
Density kg/cum	859.2045	36.96478	1.422256	2.065996	739.6681	11.11675	751.5343
Average MW	78.11364	57.95286	36.25149	2.426406	81.56542	13.90172	84.01725
Liq Vol 60F l/sec	10.20666	22.98366	0.426811	19.33138	17.99455	4.989061	12.16937
							0.610239
							4.378823

Problem B: Stream Summary - II

	RECLIQ	S1	S2	S3	S4	S5	S6	TEAR	TOPURIFY
Substream: MIXED									
Mole Flow lbmol/hr									
H2	1.260891	3022.114	263.7353	263.7353		0	227.7854	1.260891	3022.103
N2	1.26272	97.22058	97.22058	97.22058		0	81.63444	1.26272	97.22315
C1	12.8504	384.358	400.8424	400.8424		0	314.2139	12.8504	384.3795
BZ	0.543182	915.7963	1.831593	1.831593	915.2347	0.018418	0.543182	915.7889	1.267426
CH	389.7723	402.3185	1313.536	1313.536		0	12.54619	389.7723	402.3161
Mole Frac									
H2	0.003108	0.62676	0.126969	0.126969		0	0.358042	0.003108	0.626757
N2	0.003113	0.020163	0.046804	0.046804		0	0.128316	0.003113	0.020163
C1	0.031675	0.079712	0.192976	0.192976		0	0.493893	0.031675	0.079717
BZ	0.001339	0.189928	0.000882	0.000882		1	2.89E-05	0.001339	0.189926
CH	0.960765	0.083437	0.632369	0.632369		0	0.019721	0.960765	0.083437
Total Flow lbmol/hr	405.6895	4821.807	2077.166	2077.166	915.2347	636.1984	405.6895	4821.811	946.6089
Total Flow ktonne/year	131.5731	478.6446	478.6446	478.6446	284.2671	35.16644	131.5731	478.6431	307.0039
Total Flow l/sec	5.636717	525.6968	141.2022	131.0439	10.49599	95.19664	5.637323	908.53	13.15234
Temperature F	120	106.2453	156.4089	120	101.7374	134.5418	120.1522	300	120
Pressure psia	310	335	315	315	335	335	335	335	310
Vapor Frac	0	0.732202	0.356948	0.348476		0	1	0	0.997772
Liquid Frac	1	0.267798	0.643052	0.651524		1	0	1	0.002229
Solid Frac	0	0	0	0		0	0	0	0
Enthalpy J/kmol	-1.5E+08	-8326200	-1.1E+08	-1.1E+08	51211500	-3.8E+07	-1.5E+08	5567370	-1.49E+08
Enthalpy J/kg	-1823600	-333510	-1860000	-1904200	655603	-2765400	-1823300	223005	-1823600
Enthalpy Watt	-7603100	-5058500	-2.8E+07	-2.9E+07	5905590	-3081600	-7601800	3382390	-17741000
Entropy J/kmol-K	-581740	-116100	-393580	-401280	-246420	-64223.9	-581730	-79284.7	-581740
Entropy J/kg-K	-7132.23	-4650.46	-6791.34	-6924.27	-3154.65	-4619.85	-7132.03	-3175.81	-7132.225
Density kmol/cum	9.068403	1.15568	1.853501	1.997181	10.98683	0.842043	9.067427	0.668704	9.068403
Density kg/cum	739.6681	28.85186	107.4157	115.7423	858.2214	11.70585	739.5886	16.69431	739.6681
Average MW	81.56543	24.96526	57.95286	57.95286	78.11364	13.90172	81.56543	24.96516	81.56542
Liq Vol 60F l/sec	5.398365	39.31522	22.98366	22.98366	10.20666	4.378823	5.398365	39.3152	12.59618

Problem A: Block Report (Operating Conditions)

CHILLER	
Name	CHILLER

Property method	PENG-ROB
Use true species approach for electrolytes	YES
Free-water phase properties method	STEAM-TA
Specified temperature [C]	-35
Specified vapor fraction	1
Calculated pressure [bar]	1.3765645
Calculated temperature [C]	-35
Calculated vapor fraction	1
Calculated heat duty [kJ/sec]	2999.99958
Net duty [kJ/sec]	2999.99958

Compr	
Name	COMPR1
Property method	PENG-ROB
Use true species approach for electrolytes	YES
Free-water phase properties method	STEAM-TA
Specified discharge pressure [bar]	5.06625
Isentropic efficiency	0.75
Calculated brake horsepower [kW]	725.891269
Net work required [kW]	725.891269
Power loss [kW]	0
Efficiency (polytropic / isentropic) used	0.75
Calculated discharge pressure [bar]	5.06625
Calculated pressure change [bar]	3.6896855
Calculated pressure ratio	3.68035788
Outlet temperature [C]	21.7650001
Vapor fraction	1
Isentropic power requirement [kW]	544.418452
Inlet heat capacity ratio	1.16843125

Compr	
Name	COMPR2
Property method	PENG-ROB
Use true species approach for electrolytes	YES
Free-water phase properties method	STEAM-TA
Specified discharge pressure [bar]	7.852687499
Isentropic efficiency	0.75
Calculated brake horsepower [kW]	302.2339

Net work required [kW]	302.2339
Power loss [kW]	0
Efficiency (polytropic / isentropic) used	0.75
Calculated discharge pressure [bar]	7.8526875
Calculated pressure change [bar]	2.7864375
Calculated pressure ratio	1.55
Outlet temperature [C]	40.8400795
Vapor fraction	36.8392252
Isentropic power requirement [kW]	1
Inlet heat capacity ratio	1.17902757

Compr	
Name	COMPR3
Property method	PENG-ROB
Specified discharge pressure [bar]	12.0466
Isentropic efficiency	0.75
Calculated brake horsepower [kW]	359.681847
Net work required [kW]	359.681847
Power loss [kW]	0
Efficiency (polytropic / isentropic) used	0.75
Calculated discharge pressure [bar]	12.0466
Calculated pressure change [bar]	4.2966
Calculated pressure ratio	1.5544
Outlet temperature [C]	59.7260744
Vapor fraction	1

Heater	
Name	CONDENS
Property method	PENG-ROB
Specified temperature [C]	35
Specified vapor fraction	0
Calculated pressure [bar]	12.2061811
Calculated temperature [C]	35
Calculated vapor fraction	0
Calculated heat duty [kJ/sec]	-4368.62123
Net duty [kJ/sec]	-4368.62123

Flash2	
Name	SEPAR
Property method	PENG-ROB
Temperature [C]	
Pressure [bar]	5.06625
Specified heat duty [kJ/sec]	0
Outlet temperature [C]	2.19601703
Outlet pressure [bar]	5.06625
Vapor fraction	0.104690562

Flash2	
Name	SEPR2
Property method	PENG-ROB
Temperature [C]	
Pressure [bar]	7.75
Specified heat duty [kJ/sec]	0
Outlet temperature [C]	17.1300671
Outlet pressure [bar]	7.75
Vapor fraction	0.138944852

Valve	
Name	VALV1
Property method	PENG-ROB
Specified outlet pressure [bar]	7.7608866
Valve pressure specification (design mode)	P-OUT
Calculated outlet pressure [bar]	7.7608866
Calculated pressure drop [bar]	4.2857134

Valve	
Name	VALV2
Property method	PENG-ROB
Specified outlet pressure [bar]	5.07393789
Valve pressure specification (design mode)	P-OUT
Calculated outlet pressure [bar]	5.07393789
Calculated pressure drop [bar]	2.67606211

Valve	
Name	VALV3
Property method	PENG-ROB
Specified outlet pressure [bar]	1.37660145
Valve pressure specification (design mode)	P-OUT
Calculated outlet pressure [bar]	1.37660145
Calculated pressure drop [bar]	3.68964855

VII – REFERENCES:

1. “U.S. Energy Information Administration - EIA - Independent Statistics and Analysis.” *EIA*.
2. Munters, “*Psychrometric Chart*. ”
3. User, Super. “Cooling Water Prices: Utilities Cost Database.” *Intratec.us*.
4. Rashidi et al, “Exergetic optimisation of a multi-stage compression transcritical refrigeration cycle.” 2016’
5. *Chemicals, Chemicals Products, Chemicals Manufacturers, Chemicals Suppliers and Exporters Directory*.

Problem 5: Cyclohexane production(Continued)

For the cyclohexane production process of **Assignment #4**, consider the option of increasing the cyclohexane recovery by replacing the low pressure separator with a distillation column. Design the distillation column to maximize the recovery of cyclohexane and update your simulation. Recalculate your fixed capital investment and the operating costs. Update the list of all equipment (except the reactor) and recalculate the total capital cost. Update the list of all utilities and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. Start again with your own simulation from **Assignment #4**. Plant capacity is 300 KMTA of cyclohexane with plant operating data defined in **Assignment #3**. Cooling water is available by a nearby cooling tower that has been designed for dry bulb temperature of 85 deg F, relative humidity of 70% and an approach temperature of 10 deg C. The maximum hot water temperature was taken as 45 deg C for the design of the cooling tower. Multiple pressure levels of steam are available by a steam plant nearby.

REQUIRED REPORT SECTIONS

- A. Outline your steps you followed to reach your final column design
- B. Final process flowsheet printout
- C. ASPEN report for the distillation column
- D. Heat and Weight Balance for the entire unit
- E. List of all equipment for with the capital costs (see example below)
- F. List of utilities, raw materials and products with operating costs

1	2	A	B	C	D	E	F	G
	1	Cyclohexane Production Cost Estimate						
	2	EQUIPMENT LIST: CAPITAL COSTS						
	3		Project:	Assignment #2	Chemical Engineers:			
	4		Location:	Chemical Engineering Department	Date:			
	5	Equipment Class	ASPEN BlockID	Service Description	Tag	Material Cost	Installed Cost	Annual Cost, \$/yr
	6	Distillation						
	7	Towers	xxxx	Finishing Column	T-1	\$1,950,958	\$4,095,200	\$819,040
	8	Towers	xxxx	Packing, Trays, Internals	T-1			
	9	Drums						
	10	Vertical Tank	xxxx	Separation Drum	V-1			
	11	Vertical Tank	xxxx	Reflux Drum	V-2			
	12	Heat Exchangers						
	13	S&T Exchanger		Feed-Product Exchanger	E-1			
	14	S&T Exchanger		Product Cooler	E-2			
	15	Air Coolers						
	16	Air Cooler		Reflux Condenser	EA-1			
	17	Air Cooler			EA-2			
	18	Pumps						
	19	Pump		Reflux Pump	P-1			
	20	Pump			P-2			
	21	Turbines						
	24	Compressors						
	27	Reactors						
	30							

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I - EXECUTIVE SUMMARY:

B. Cyclohexane Production

The cyclohexane production schematic from previous homework was reconfigured using the real-life equipment. The objective of this homework is to 1) increase the recovery of cyclohexane via hydrogenation of benzene in bubble column reactor in the presence of nickel catalyst by keeping the plant capacity 300 KMTA and 2) Recalculate the fixed capital investment, total operating costs, and total capital costs. The increase of recovery of cyclohexane will be achieved using three major steps which is briefly discussed in this section. To achieve the maximum recovery of the cyclohexane first, the TOPURIFY stream exiting from the high-pressure separator is introduced in SEP2, a two-product separator unit, to compute the recovery and purity of the cyclohexane. The advantage of using SEP2 is that we can specify the split fractions required in the product streams. The optimal pressure at which the recovery of cyclohexane is 0.9999 was found to be 1.03 bar (~15 psia). The found pressure was then utilized to simulate DISTWU, a shortcut model, to compute actual reflux ratio, number of stages, feed stage, and distillate to feed ratio, which was found to be 0.37, 10, 6, and 0.038 respectively. The information found from the DISTWU model is then used to simulate RADFRAC, a rigorous model, to finally compute maximum recovery of cyclohexane which was then found to be 0.9986 at the production rate of 300 KMTA. Unlike shortcut model, rigorous model calculations are accurate since the model performs stage by stage calculations, does not consider ideal behavior between liquid and gas phases, allow multiple types of design specifications, and allow multiple column configurations.

The list of equipment will be slightly modified since now we will be using a reflux pump and a distillation column in addition to a shell and tube heat exchanger system, a cooler, two pumps and a compressor. After running the simulation and doing cost analysis, the total capital cost was found to be \$4,589,103. Total operating cost and the cost of raw materials for the utilities was found to be \$116,114.09 and \$216,563,328.00 respectively.

II – SYSTEMS AND PROCEDURES:

PROBLEM A:

To run the simulation, two **Design Specifications** were defined. Using the data from Assignment #3, for the tear stream, for the present composition of hydrogen and benzene in the tear namely TEARH and TEARB. In the **Spec** tab the ratio of TEARH and TEARB was targeted to be equal to 3.3 by **Varying** the flow fraction of the splitter B4. **Second design specification** was defined for the product stream (PRODUCT) where the mass flow rate of the product was targeted to be equal to **300 KMTA** by **Varying** the benzene feed (BZFEED) molar flow rate. The molar flow of hydrogen was reevaluated after the design spec was set up using the **Calculator** in ASPEN. Since the recycle liquid should not be more than 45 % of the benzene feed, the ratio of liquid recycles (RECLIQ) and the benzene feed (FEEDB) was made equal to 0.45 again using the **Calculator**. New variables **FEEDB, HNEW, and RECLIQ** were defined in order to make the Fortran calculation.

In order to replace the low-pressure separator with a rigorous distillation column the following steps must be taken:

1. SEP2 Model: To achieve the maximum recovery of the cyclohexane, TOPURIFY stream exiting from the high-pressure separator is introduced in SEP2, a two-product separator unit, to compute the recovery and purity of the cyclohexane. The required split fractions for the product was specified. After running the simulation, the optimal pressure to achieve the maximum recovery was noted.

2. DISTWU Model: The pressure obtained from the SEP2 simulation is then utilized to run DISTWU, a shortcut model which determines the minimum and actual reflux ratio, minimum and actual number of trays, and optimal feed tray location. In DISTWU model, we assumed the reboiler pressure equals to the feed pressure (15 psia) and the pressure drop across the column to be 1 psia. The reflux ratio was defined to be $R = -1.15 R_{MIN}$. Condenser was specified as partial condenser with vapor distillate since the fuel stream is exiting the top of the column in the vapor phase. Actual reflux ratio, number of stages, feed stage, and distillate to feed ratio was determined after plugging in the above information.

3. RADFRAC Model: The information found from the DISTWU model is then utilized to simulate RADFRAC, a rigorous model, to compute maximum recovery of cyclohexane. Again, partial vapor condenser was selected in ASPEN and the condenser utility was specified to be cooling water. Feed location, reflux ratio, and distillate to feed ratio was specified to be stage #6, 0.37, and 0.038 respectively using the data from DISTWU model.

The simulation was then ran using RADFRAC and the recovery of cyclohexane was then found to be 0.9986 at the production rate of 300 KMTA.

III – RESULTS:

PROBLEM A:

Table 1: Equipment used for cyclohexane production with the capital costs

Cyclohexane Production Cost Estimate				
EQUIPMENT LIST: CAPITAL COSTS				
Equipment Class	Assignment #5			
	Chemical Engineering Department			
	Service Description	Material Cost (Equipment Cost)	Installed Cost	Annual Cost, \$/yr
Drums				
Vertical Tank	High Pressure Drum	\$52,855	\$39,854	\$53,138
Heat Exchangers				
S&T Exchanger	Feed-Product Exchanger	\$228,084	\$798,295	\$1,064,393
Water Coolers				
Water Coolers	Reflux Condenser	\$66,702	\$233,458	\$311,277
Reflux Pumps				
Pump1	Pump used to pressurize benzene feed	\$9,963	\$39,854	\$53,138
Pump2	Pump used to recycle liquid	\$9,122	\$36,488	\$48,651
Pump3	Reflux Pump			
Compressors				
Compressor	Recycle gas compressor	\$707,895	\$1,769,737	\$2,359,649
Distillation Column				
RADFRAC	Separation Column	\$13,802	\$55,208	\$73,610.87
Condenser	Condenser	\$66,702	\$233,458	\$311,277.43
Reboiler	Reboiler	\$66,702	\$233,458	\$311,277.43
Trays	Trays	\$505	\$2,018	\$2,690.79

Table 2: Utilities, Raw Material, Product with operating costs

A	Utility Class						
1	ELECTRICITY	Power (kW)	Cost(\$/kW)	Hours	Yearly Cost (\$)		
			0.09				
	Pump1	30.361934	0.09	8640	23609.44003		
	Pump2	1.2753211	0.09	8640	991.6897107		
	REFLUX Pump3	0.02387	0.09	8640	18.56127957		
	Compressor	22.031155	0.09	8640	17131.42589		
	Total				41751.11692		
2	Cooling Water	Power (kW)	\$/kg	LMTD (K)	Specific Heat Capacity(kJ/kg.s)	Yearly Cost (\$)	
	(Condenser)	94.21036	0.02113	201.625324	4.187	74362.96992	
B	Raw Materials						
		\$/MT	MT/hr	hR/YR	\$/YR		
	BENZENE	760	31.92	8640	209599488		
		\$/MT	MT/HR	HR/YR	\$/YR		
	HYDROGEN	260	3.1	8640	6963840		
C	PRODUCT						
		\$/TON	TON/YR		\$/YR		
	CYCLOHEXANE	1040	300000		312000000		

Table 3: Raw Materials and Product

Parameters	Value	Units
Product	300	ktonne/year
Benzene Feed	901.007	lbmol/hr
Hydrogen Feed	2820.15	lbmol/hr
Cyclohexane Recovery	0.9999	

Table 4: Total Costs

Total Capital Cost	Total Operating Cost	Total Raw Material Cost	Product Revenue
\$4,589,103	\$116,114.09	\$216,563,328.00	\$312,000,000.00

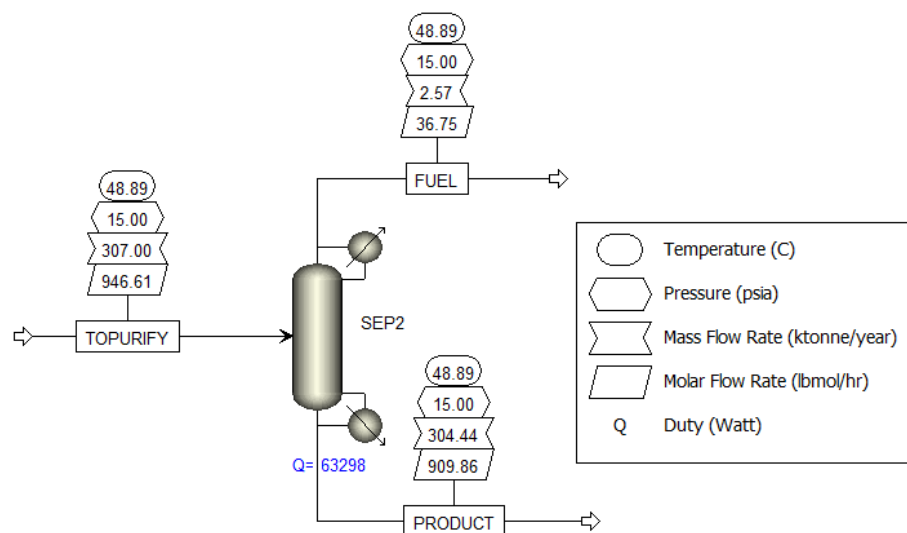


Figure 1: Results of the SEP2 simulation

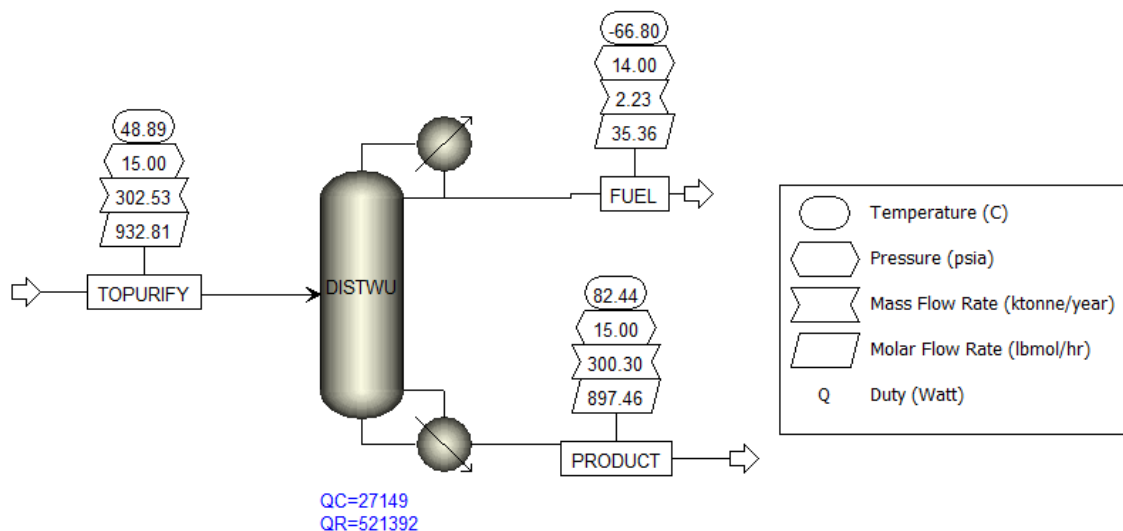


Figure 2: Results of the DISTWU simulation

V – CONCLUSIONS:

PROBLEM A: The aim of this project was to simulate the actual production plant to produce cyclohexane using the bubble column reactor. By focusing on the accurate utilities, such as, rigorous distillation column, a shell and tube heat exchanger system, a cooler, reflux pumps and a compressor were installed to maximize the recovery of cyclohexane. After replacing a low-pressure separator with a distillation column, the recovery of cyclohexane was increased from 98 % to 99.86 %. In addition, the total operating costs, total capital costs, and fixed capital investments were recalculated by taking additional equipment into account. The total capital cost was calculated by utilizing the equations that calculate the cost of the equipment by taking in account of the indexes of the year 2010. The value of the total capital cost was found to be \$4,589,103. The total operating cost was also evaluated utilizing the annual cost required for the maintenance of the equipment which was found to be \$116,114.09. The total cost of the raw materials was evaluated to find out how much money should be spent on buying raw materials to produce 300 KMTA cyclohexane. The raw material cost was found to be \$216,563,328.00.

VI – APPENDICES:

APPENDIX I:

PROBLEM A:

For Design Specifications in ASPEN Plus we used,

$$\frac{TEARH}{TEARB} = 3.3$$

$$HNEW = 3.13 * FEEDB \text{ (SOLVER ASPEN)}$$

Calculating equipment, installation, and annualized cost

For Shell and Tube Heat Exchanger, Cooler and Condenser:

Utilizing the cost data for the year 2010 we have,

$$\$ (2010) = 28000 + 54 (A[m^2])^{1.2}$$

Where A can be found using,

$$Q = F_T A U \Delta T_{LN}$$

Where, ΔT_{LN} is given as,

$$\Delta T_{LN} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

The installation factor for the shell and tube heat exchanger is given to be 3.5 therefore,

$$\text{Installation cost} = 3.5 * \text{purchasing cost}$$

The annualized cost can be found using,

$$\text{Annualized cost} = \frac{4}{3} * \text{installation cost}$$

For Reflux Pumps

$$\$ (2010) = 8000 + 240 \left(Q \left[\frac{L}{s} \right] \right)^{0.9}$$

Installation factor is given to be 4.0. Also, the installation cost and annualized cost can be found from the same formulae.

The increased in pressure is given as,

$$\Delta P = \rho g h$$

Where, ΔP is the increase in pressure, g is force of gravity and h is the height of the column.

For Compressor

$$\$ (2010) = 58000 + 20000 (W[kW])^{0.6}$$

Installation factor is given to be 2.5. Also, the installation cost and annualized cost can be found from the same formulae.

For Drums (Vertical Vessel)

$$\$ (2010) = 11600 + 34 (M[kg])^{0.85}$$

Where M is given as,

$$M[kg] = \pi D_i L t \rho_{STEEL}$$

Where t is the thickness of the walls of the drum and is given as,

$$t = \frac{P_i D_i}{2 S E - 1.2 P_i}$$

Where,

D_i is the internal diameter

P_i is the internal pressure of the drum + 10 % safety factor

E = welded efficiency

S = Stress material can take [=] ksi

Where,

$$I \text{ ksi} = 6.895 \frac{N}{m^2}$$

Internal Diameter can be calculated using the residence time,

$$\tau = \frac{V}{F}$$

Where volume of the vessel is given as,

$$V_{VESSEL} = 2 V$$

And the length is given as,

$$L = 2 D_i$$

Where,

$$D_i = \left(\frac{2 V_{VESSEL}}{\pi} \right)^{\frac{1}{3}}$$

Installation factor is given to be 4.0. Also, the installation cost and annualized cost can be found from the same formulae.

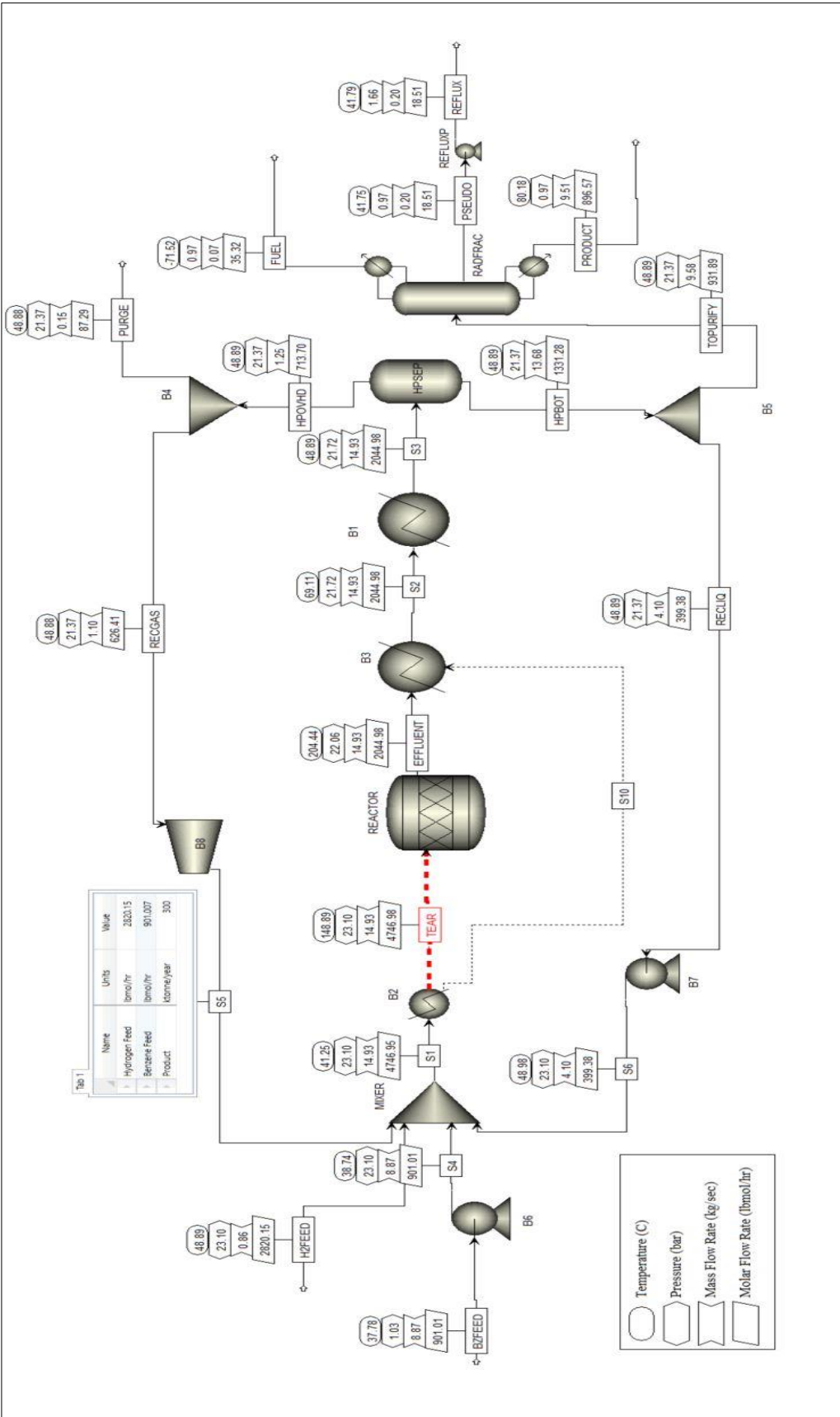
Table 5: Standard Prices for 2010 Used for Cost Estimation

Utility Class	Prices	Units
Electricity	0.09	USD/KWh
High-Pressure Steam	11.45	USD/1000 lb
Medium-Pressure Steam	9.50	USD/1000 lb
Low-Pressure Steam	8.48	USD/1000 lb
Cooling Water	0.08	USD/1000 gal
Fuel Gas (LHV)	7.30	USD/MMBtu
Raw Materials		
Benzene	760	USD/MT
Hydrogen	2.60	USD/MSCF
Product		
Cyclohexane	1040	USD/MT

Table 6: Installation Factors Used for Calculations

Equipment	Installation Factor
Compressor	2.5
Distillation Column	4.0
Heat Exchanger	3.5
Pressure Vessels	4.0
Pumps	4.0
Fired Heater	2.0
Other	2.5

Figure 3: Cyclohexane Production Process Flow Diagram



APPENDIX II:

APPENDIX III:

Problem A: Stream Summary

	BZFEED	EFFLUENT FUEL	H2FEED	HPBOT	HPOVHD	PRODUCT	PURGE	REGAS	PSEUDO
Substream: MIXED									
Mole Flow lbmol/hr									
H2	0	259.6648	2.89626	4.137514	255.5248	1.90E-13	31.25104	224.2737	0.0004081
N2	0	95.72174	2.900504	4.143578	91.5781	5.41E-11	11.20013	80.37797	0.0012274
C1	0	394.6713	29.51892	42.16988	352.501	1.22E-07	43.11137	309.3896	0.0351042
BZ	901.007	1.80312	1.05E-05	0	1.782458	0.020662	1.24771	0.00252705	0.0181354
CH	0	1293.117	0.005423	0	1279.042	14.07376	895.3241	1.72124	12.35252
Mole Frac									
H2	0	0.1269768	0.081998	0.975	0.003108	0.358029	2.12E-16	0.3580291	0.3580291
N2	0	0.0468082	0.082118	0.005	0.003112	0.128315	6.03E-14	0.1283149	0.1283149
C1	0	0.1929954	0.83573	0.02	0.031676	0.493908	1.36E-10	0.4939076	0.4939076
BZ	1	0.0008817	2.97E-07	0	0.001339	2.90E-05	0.0013917	2.90E-05	2.90E-05
CH	0	0.6323379	0.000154	0	0.960764	0.019719	0.9986084	0.0197194	0.0197194
Total Flow lbmol/hr	901.007	2044.978	35.32111	2820.152	1331.276	713.6983	896.5719	87.28631	626.4119
Total Flow									
ktonne/year	279.848	471.2096	2.231096	431.7587	39.45056	300	4.824859	34.6257	6.1834803
Total Flow cum/sec	0.010321	0.4039668	0.076927	0.41732	0.018497	0.0132149	0.0137526	0.0986959	0.0002593
Temperature F	100	400	-96.7436	120	120	176.3218	119.9842	119.9842	314.90422
Pressure psia	15	320	14	335	310	310	14	310	96526.602
Vapor Frac	0	1	1	1	0	1	0	0.9999946	0.9999946
Liquid Frac	1	0	0	0	1	0	1	5.38E-06	5.38E-06
Solid Frac	0	0	0	0	0	0	0	0	0
Enthalpy J/kmol	50944100	-75538000	-6.6E+07	-777550	-1.5E+08	-3.9E+07	-1.47E+08	-38724000	-153115228
Enthalpy J/kg	652179	-1303500	-4143600	-320450	-1823600	-2785500	-1747300	-2785500	-2785500
Enthalpy Watt	5783420	-19463000	-292950	-276290	-2.5E+07	-3482200	-16611000	-425880	-3056300
Entropy J/kmol-K	-246580	-316790	-75505.1	-24322.7	-581740	-64436.8	-585770	-64438.07	-603444.68
Entropy J/kg-K	-3156.665	-5466.546	-4752.91	-10024.2	-7132.23	-4635.14	-6960.795	-4635.229	-4635.229
Density kmol/cum	10.99942	0.6378318	0.057852	0.851464	9.068406	0.799671	8.548393	0.799694	0.799694
Density kg/cum	859.2045	36.96277	0.919047	2.065996	739.668	11.11687	719.3718	11.11719	11.11719
Average MW	78.11364	57.95065	15.88608	2.426406	81.56538	13.90181	84.15286	13.90181	13.90181

Problem A: Stream Summary

	RECLIQ	S1	S2	S3	S4	S5	S6	TEAR	TOPURIFY	REFLUX
Substream: MIXED										
Mole Flow lbmol/hr										
H2	1.241311	2975.163	259.6648	259.6648	259.6648	0	224.2737	1.241311	2.89626	0.0004081
N2	1.243009	95.72174	95.72174	95.72174	95.72174	0	80.37797	1.243009	2.900504	0.0012274
C1	12.65055	378.4432	394.6713	394.6713	394.6713	0	309.3896	12.65055	29.51892	0.0351042
BZ	0.5347385	901.5598	1.80312	1.80312	1.80312	901.007	0.018135	0.5347385	1.247721	0.0335755
CH	383.7124	396.0649	1293.117	1293.117	1293.117	0	12.35252	383.7124	895.3296	18.439663
Mole Frac										
H2	0.00310808	0.6267522	0.126977	0.126977	0.126977	0	0.358029	0.0031081	0.00310793	2.21E-05
N2	0.00311233	0.0201648	0.046808	0.046808	0.046808	0	0.128315	0.0031123	0.00311249	6.63E-05
C1	0.0316753	0.0797233	0.192995	0.192995	0.192995	0	0.493908	0.0316753	0.0316762	0.0018965
BZ	0.00133892	0.1899239	0.000882	0.000882	0.000882	1	2.90E-05	0.0013389	0.00133891	0.0018139
CH	0.9607654	0.0834356	0.632338	0.632338	0.632338	0	0.019719	0.9607654	0.9607644	0.9962012
Total Flow lbmol/hr	399.382	4746.953	2044.978	2044.978	2044.978	901.007	626.4119	399.382	931.893	18.509978
Total Flow ktonne/year	129.5275	471.2096	471.2096	471.2096	471.2096	279.848	34.6257	129.5275	302.2311	6.1834803
Total Flow cum/sec	0.00554908	0.5175386	0.139023	0.129023	0.10333	0.093728	0.0055497	0.8944383	0.0129478	0.0002593
Temperature F	120	106.2444	156.4024	120	101.7374	134.5179	120.1551	300	120	314.93658
Pressure psia	310	335	315	315	315	335	335	335	310	165569.33
Vapor Frac	0	0.7322073	0.356979	0.348509		0	1	0	0.9977804	0
Liquid Frac	1	0.2677927	0.643021	0.651491		1	0	1	0.0022196	1
Solid Frac	0	0	0	0	0	0	0	0	0	0
Enthalpy J/kmol	-148740000	-8327000	-1.1E+08	-1.1E+08	51211500	-3.8E+07	-1.49E+08	5566800	-148740000	-153104993
Enthalpy J/kg	-1823600	-333550	-1860000	-1904200	655603	-2765500	-1823300	222985	-1823600	-1822340.3
Enthalpy Watt	-7484900	-4980400	-2.8E+07	-2.8E+07	5813790	-3034300	-7483600	3329560	-17465000	
Entropy J/kmol-K	-581740	-116100	-393560	-401260	-246420	-64226.1	-581730	-79283.54	-581740	-603438.4
Entropy J/kg-K	-7132.225	-4650.46	-6791.34	-6924.25	-3154.65	-4619.98	-7132.025	-3175.794	-7132.225	-7182.4576
Density kmol/cum	9.068399	1.155674	1.85339	1.997034	10.98683	0.842079	9.067407	0.6686986	9.068406	8.9940806
Density kg/cum	739.668	28.85144	107.4052	115.7294	858.2214	11.70642	739.5871	16.69403	739.668	755.64297
Average MW	81.56544	24.96502	57.95065	57.95065	78.11364	13.90181	81.56544	24.96495	81.56538	84.015589

Problem A: Block Summary (Operating Conditions)

Pump1			PUMP2			COMPRESSOR (ISENTROPIC)	
Outlet Pressure (psia)	335		Outlet Pressure (psia)	335		Discharge Pressure (psia)	335
Pump Efficiency	0.75		Pump Efficiency	0.75		Compr. Efficiency	0.75
COOLER			HEATER1			HEATER2	
TEMP-OUT (F)	120		TEMP (F)	300		TEMP (F)	156.41
Outlet Pressure (psia)	315		Outlet Pressure (psia)	335		Outlet Pressure (psia)	315
Net Duty (Watt)	-659283		Net Duty (Watt)	8310057.49		Net Duty (Watt)	0
REACTOR			HSEP				
TEMP (F)	400		TEMP (F)	120			
Outlet Pressure (psia)	320		Outlet Pressure (psia)	310			
RADFRAC			Reboiler/Bottom Stage performance				
Condenser/Top Stage Performance			Reboiler/Bottom Stage performance				
Temperature	-96.7436	F	Temperature	176.3	F		
Heat duty	-94211	Watt	Heat duty	7E+05	Watt		
Distillate rate	35.32111	lbmol/hr	Bottoms rate	896.6	lbmol/hr		
Reflux rate	13.15146	lbmol/hr	Boilup rate	171.6	lbmol/hr		
Reflux ratio	0.37234		Boilup ratio	0.191			
Distillate to feed ratio	0.037903		Bottoms to feed ratio	0.962			

REFLUX PUMP	
Outlet Pressure (psia)	10.0138
Pump Efficiency	0.75

VII – REFERENCES:

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